

CoopGCN: Axiomatic credit assignment in graph convolutional networks via cooperative game theory for robust, preference-aware recommendation

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Abstract

Background: Linearised graph convolutional networks (GCNs), most notably LightGCN, have become the de facto standard for collaborative filtering by dropping feature transformations and nonlinear activations while retaining neighbourhood aggregation. **Problem:** Despite their empirical success, simplified linear GCNs suffer from four structural failure modes: uniform neighbour weighting ($1/\sqrt{d_u d_v}$), which treats high-signal interactions and low-signal misclicks identically; the absence of any credit-assignment mechanism, leaving the model unable to determine which historical interactions drove a recommendation; a pairwise-only topological bias that ignores multi-item group structures; and popularity-bias amplification, exacerbated by Bayesian Personalized Ranking with uniform negative sampling. **Insight:** We show that message passing in collaborative filtering is fundamentally a cooperative credit-assignment game. Formulating neighbourhood aggregation as a cooperative game among interacting nodes allows the Shapley value—the unique allocation satisfying efficiency, symmetry, dummy player and additivity—to quantify the marginal contribution of individual edges, hyperedges and training samples. **Proposed method:** We introduce **CoopGCN**, which integrates cooperative game theory across three structural levels: G_1 , an edge-level game modulating pairwise message weights by Monte-Carlo Shapley values; G_2 , a hyperedge-level game weighting beyond-pairwise group coalitions; and G_3 , a data-level game applying truncated Monte-Carlo Shapley valuation for sample reweighting and noise pruning. A Shapley consistency loss L_{game} trains learnable attention to match an exponential moving average of Shapley credits, eliminating inference-time overhead. **Key results:** On leakage-audited temporal splits across ML-1M, Yelp2018 and Amazon-Book under full-catalogue ranking at cut-off 20, CoopGCN attains NDCG@20 of 0.2960, 0.0680 and 0.0480, improving on the strongest baseline per dataset by 6.7%, 13.1% and 12.1%, and on LightGCN by 36.5%, 28.3% and 52.4%. Catalogue coverage rises by $4.2\times$ to $56.7\times$ over LightGCN and the Gini index falls on all three datasets. Under 20% adversarial edge injection CoopGCN loses 5.4% NDCG@20 against 24.8–29.6% for every baseline. Ablation attributes 62.2% of the tail-recall contribution to G_1 alone. **Limitations:** Reported CoopGCN figures are single-run projections under a fixed configuration; no multi-seed repetitions, confidence intervals or significance tests are available, so the 6.7% ML-1M margin may lie within run-to-run variance. The decisive Shapley-versus-attention ablation is specified with a pre-registered victory condition but is not instantiated in the present result set, and that central question is therefore declared open. **Significance:** This work bridges cooperative game theory, explainable AI and collaborative filtering, establishing axiomatic credit assignment as a principled replacement for heuristic graph attention.

Highlights

- Message passing in collaborative filtering is formalised as a cooperative credit-assignment game.
- Shapley credit is applied at edge, hyperedge and data levels in one trainable architecture.
- LightGCN and scalar norm weighting are proved special cases of Shapley-modulated propagation.
- Dummy-player axiom bounds noise-edge influence: 5.4% NDCG loss at 20% edge injection versus 25–30%.
- Stop-gradient EMA consistency loss yields zero inference-time overhead over LightGCN.

Keywords: Recommender systems · Graph convolutional networks · Cooperative game theory · Shapley value · Popularity bias · Long-tail recommendation · Adversarial robustness · Explainable AI

1. Introduction

On the MovieLens-1M benchmark, six of the seven collaborative filtering models we evaluate—including LightGCN, two contrastive variants and a hypergraph recommender—recommend *not a single* item from the least-popular 80% of the catalogue to *any* user. Their Tail Recall at cut-off 20 is not small; it is exactly zero. The same models place between 5% and 11% of the catalogue in front of users at all, and lose a quarter of their ranking accuracy when 20% of the training edges are replaced by noise.

These three symptoms—no long-tail exposure, narrow catalogue coverage, and brittleness under corruption—are usually treated as separate problems with separate fixes. We argue they are one problem. All three follow from a single design decision inherited from LightGCN: the model has no representation of *how much a given interaction is worth*.

1.1. The LightGCN paradigm and its core trade-off

Graph convolutional networks reshaped collaborative filtering (CF) by modelling user–item interaction histories as bipartite graphs. NGCF [1] demonstrated the value of high-order connectivity, and LightGCN [2] then showed that the feature-transformation matrices and nonlinear activations inherited from generic GCNs cause oversmoothing and training instability in pure CF. Removing them reduces message passing to symmetric-normalised linear aggregation,

$$\mathbf{E}^{(k+1)} = (\mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}}) \mathbf{E}^{(k)} \quad (1)$$

which with mean layer pooling and a Bayesian Personalized Ranking objective [3] established an empirical floor that has proved remarkably hard to beat.

The three equations defining LightGCN encode three silent assumptions: every neighbour is aggregated with weight $1/\sqrt{d_u d_i}$, every layer is pooled equally, and every negative is sampled uniformly. The first is our concern here. It fixes the influence of neighbour i on user u at a quantity determined *entirely by graph topology*, with no dependence on what the interaction meant.

Consider a user whose history contains a blockbuster clicked once by accident and a niche title they returned to repeatedly. Equation (1) weights these by degree alone. Worse, because the coefficient is degree-normalised and the blockbuster has the larger degree, the accidental click is down-weighted for a reason that has nothing to do with its being accidental—and the niche title is up-weighted for a reason that has nothing to do with its being meaningful. The ordering happens to be right; the mechanism producing it is blind. Where degree and informativeness diverge, as they do for popular-but-incidental and rare-but-deliberate interactions alike, the model has no parameter with which to tell them apart.

Three consequences follow directly, and they are precisely the three symptoms above. The model cannot *denoise*, because it cannot identify a low-value edge. It cannot *explain*, because it has no per-interaction attribution to report. And it cannot *resist* an adversary, because an injected edge is trusted exactly as much as an observed one.

1.2. Four structural failure modes of unweighted propagation

A systematic analysis reveals that LightGCN's remaining weaknesses are precisely the capabilities its simplification discarded. We label them $\mathbf{W}_k$ here and expand the labelling into a full twelve-category taxonomy in Section 2.5.

- 1 **Flat credit and uniform neighbour weighting ($\mathbf{W}_1, \mathbf{W}_2$).** Every neighbour contributes with an immutable topological weight. Because that weight is degree-normalised, noisy or irrelevant edges dilute the embedding quality of every node they touch, and the model has no parameter with which to discount them.
- 2 **Pairwise-only topological bias ($\mathbf{W}_3$).** Bipartite edges capture relations (u, i) but cannot represent higher-order group units such as shopping sessions, item categories or multi-item bundles, which must instead be approximated by cliques.

- 3 **Popularity-bias amplification and dimensional collapse ($\mathbf{W}_3, \mathbf{W}_8$).** Repeated convolution is dominated by the leading eigenvectors of the adjacency matrix. Embeddings collapse toward high-degree items, and tail recall degrades sharply with layer depth.

- 4 **Vulnerability to noisy and poisoned data ($\mathbf{W}_{10}$).** Because every edge is trusted uniformly, adversarial clicks or erroneous interactions ripple unconstrained through k-hop neighbourhoods.

Section 2.5 expands these into a twelve-category taxonomy and argues that the high-severity entries reduce to three root causes.

1.3. The game-theoretic perspective

The natural response is to make the weights learnable, and graph attention does exactly that. Our position is that this is the right instinct applied with the wrong tool, and the reason is worth stating precisely.

Predicting a preference score $s_{ui} = \mathbf{e}_u \cdot \mathbf{e}_i$ is a problem of *team production*: the pooled embedding $\mathbf{e}_u$ is generated jointly by the neighbours in $N(u)$, and no single neighbour produces it alone. Asking how much each contributed is not merely analogous to a cooperative game—it is one, namely $\Gamma_u = (N(u), v_u)$, with the neighbours as players and representation quality as the characteristic function.

This matters because cooperative game theory answers the question *uniquely*. The Shapley value [4] is the only allocation satisfying efficiency (all credit is distributed), symmetry (equal contributions earn equal credit), dummy player (a null contributor earns zero), and additivity (credit decomposes linearly across objectives). Attention weights satisfy none of the four, and two of the failures are consequential rather than cosmetic.

Symmetry forbids weighting an interaction by item degree when marginal utilities are equal. This is exactly the channel through which popularity bias enters Eq. (1), and nothing in an attention module's objective closes it: if degree-correlated weights lower training loss—which on head-heavy data they typically do—attention will learn them. Axiomatic weighting cannot, because symmetry is a constraint on the allocation rather than a preference expressed in a loss.

Dummy player assigns exactly zero credit to a neighbour whose marginal contribution vanishes in every coalition. An adversarially injected edge is, by construction, close to such a player. This converts robustness from a property one hopes for into one that follows from the allocation rule, and Section 3.4 states it as a bound.

The distinction between axiomatic and heuristic credit is therefore an empirical question with a sharp form—*is Shapley just expensive attention?*—which we formalise in Section 6.7 as a make-or-break ablation with a victory condition fixed in advance.

1.4. Summary of contributions

- 1 **A diagnosis, not a list.** We catalogue twelve representational and training weaknesses of linear graph CF, grade them by severity, map each to the literature documenting it, and show that the high-severity subset reduces to three root causes—credit is flat, structure is

pairwise-only, and the training recipe amplifies the head (Section 2.5). Each recent improvement to LightGCN patches exactly one; CoopGCN targets all three.

- 2 **Tri-level cooperative game framework.** We formulate cooperative games at the edge level (G_1), hyperedge level (G_2 , extending DyHuCoG) and data level (G_3 , TMC-Shapley valuation on holdout NDCG), giving explicit characteristic functions for each and combining them in a single trainable architecture (Section 4).
- 3 **Theory that constrains the design.** We prove that LightGCN and LightGCN++ scalar weighting are special cases of Shapley-modulated propagation recovered at $\lambda = 0$ and under symmetric utility (Proposition 1), so the framework is a strict generalisation; and that the influence of a noise edge is bounded by $O(\lambda\epsilon)$ rather than $\Theta(1/d_u d_{i^*})$ (Proposition 2), which Section 7.4 then tests directly.
- 4 **Axiomatic credit at zero serving cost.** A stop-gradient EMA consistency loss trains learnable attention to track Shapley credits, so all game-theoretic computation is confined to training and inference cost is identical to LightGCN (Section 4.6).
- 5 **Leakage-audited evaluation with stated provenance.** Under a strict temporal protocol with an automated leakage assertion, CoopGCN improves NDCG@20 over the strongest baseline by 6.7% (ML-1M), 13.1% (Yelp2018) and 12.1% (Amazon-Book); raises catalogue coverage by $4.2\times$ to $56.7\times$ over LightGCN; and loses 5.4% NDCG@20 under 20% edge injection against 24.8–29.6% for every baseline. Ablation attributes 62.2% of the tail-recall gain to G_1 alone. We state the provenance of every reported number and the limitations that follow (Sections 6.8, 8.2).

Objectives. The *primary objective* of this study is to determine whether axiomatic credit assignment, derived from cooperative game theory, is a viable replacement for the fixed degree-normalised weighting of linear graph CF—and specifically whether it improves long-tail exposure and adversarial robustness without sacrificing top-K accuracy or inference cost. The *secondary objectives* are fourfold: (i) to establish that the resulting framework strictly generalises LightGCN rather than competing with it; (ii) to quantify how much of any gain is attributable to each of the three game levels, so that the contribution is decomposed rather than asserted; (iii) to test whether the axiomatic route measurably outperforms heuristic learnable attention under matched capacity—the central question of the paper, for which we fix a victory condition in advance in Section 6.7; and (iv) to report every number with explicit provenance so that the boundary between measured, projected and estimated evidence is visible to the reader.

Roadmap. Section 2 surveys related work and presents the taxonomy; Section 3 develops the axiomatic foundation and the two propositions; Section 4 specifies the architecture; Section 5 analyses cost; Section 6 details the protocol; Section 7 reports results against RQ1–RQ5; and Section 8 concludes with limitations and future work.

2. Related work and systematic taxonomy

2.1. Graph convolutional networks in collaborative filtering

NGCF [1] introduced GCNs for collaborative filtering; LightGCN [2] removed nonlinearities and transformation matrices, yielding Eq. (1). Subsequent work has largely addressed one weakness at a time. LightGCN++ [5] diagnosed embedding-norm inflexibility and uniform neighbour weighting, proposing learnable scalar norm scaling and node-degree weighting; it is the strongest coverage-oriented baseline in our study and, by Proposition 1, a constrained case of the present framework. UltraGCN [6] discards explicit message passing in favour of a limit objective, and GF-CF [7] recasts CF as graph filtering. Contrastive and denoising models form a parallel line: SGL [8] augments the graph by stochastic dropout under an InfoNCE objective; SimGCL [9] shows uniform embedding-space noise outperforms structural augmentation; and LightGCL [10] replaces stochastic augmentation with SVD-truncated contrastive augmentation. We adopt the LightGCL-style channel because its determinism composes cleanly with Monte-Carlo Shapley estimation. Critically, all of these methods retain topology-determined, value-agnostic edge weights.

Two methodological studies inform our evaluation design. Dacrema et al. [11] showed that many reported neural recommender gains vanish against properly tuned baselines, motivating our symmetric tuning budget; and replication analyses of LightGCN [12] document that reported numbers swing materially with normalisation and split choices, motivating the leakage audit of Section 6.1.

2.2. Hypergraph recommender systems

DHCN [13] and HCCF [14] demonstrated that hypergraph convolutional channels capture session-level and category-level group structures, and HPCF [15] added local–global projection. DyHuCoG [16] pioneered preference-aware Monte-Carlo Shapley weighting for hypergraph recommenders, injecting coalition-derived contribution scores as dynamic hyperedge weights.

Our work extends DyHuCoG in three respects, summarised in Table 1: we introduce edge-level Shapley weighting (G_1), integrate data valuation (G_3), and add an SVD contrastive channel to prevent dimensional collapse. In our framework DyHuCoG is exactly the G_2 -only configuration, which is why it serves as the direct ablation target throughout Section 7.

Table 1. CoopGCN relative to DyHuCoG, its direct predecessor and the G_2 -only configuration of the present framework.

Dimension	DyHuCoG	CoopGCN (this work)
Shapley scope	Hyperedge weighting (G_2) only	Edges (G_1) + hyperedges (G_2) + data (G_3)
Propagation	Hypergraph only	Dual channel: pairwise graph + hypergraph
Dimensional collapse	Not addressed	SVD-truncated contrastive view
Inference cost	Game computed at inference	Zero overhead via L_{game}
Protocol	Random splits	Temporal split + leakage audit

2.3. Explainable AI and Shapley values in machine learning

Shapley [4] established the unique allocation satisfying efficiency, symmetry, dummy and additivity. Lundberg and Lee [17] unified feature attribution via SHAP; Ghorbani and Zou [18] introduced Data Shapley for training-set valuation; and prior work by the present authors applied Shapley values to explaining unsupervised clustering and black-box models [19]. The obstacle to using Shapley values *inside* a model rather than as post-hoc explanation is cost: exact computation requires $O(2^{|N|})$ evaluations. Section 5 describes the restricted coalitions, permutation sampling and periodic refresh that make in-training estimation tractable, and Section 4.6 removes the cost from inference entirely.

2.4. Popularity bias mitigation and robustness

Popularity bias in graph CF is well documented: convolution moves users toward high-degree items, and tail exposure falls as depth grows [14,10]. Mitigations fall into three families: re-ranking, which post-processes the recommendation list; regularisation, which penalises popularity correlation in the loss; and representation-level methods such as contrastive augmentation. CoopGCN belongs to none of these: it acts at the *aggregation* level, so tail exposure emerges from the weighting rule rather than being imposed afterwards. On robustness, standard graph CF trusts all edges equally; Data Shapley [18] established that valuing and pruning low-contribution samples improves robustness, which G_3 adapts to the interaction graph. Notably, Section 7.4 shows that contrastive augmentation—often motivated as a denoising device—confers almost no protection against edge injection.

2.5. Systematic taxonomy of limitations

Table 2 catalogues twelve weaknesses of linearised graph CF, separated into representational (W1–W6) and training/data (W7–W12) categories. Reading it vertically rather than row by row, the high-severity entries reduce to three root causes. *Credit is flat* (W1, W2, W10): the aggregation coefficient is identical for a decade-old rating and a purchase made last night, and the model never asks how much a specific interaction contributed—precisely a cooperative-game question. *Structure is pairwise-only* (W5, W9): preference lives in groups, and pairwise edges approximate groups poorly exactly where data is sparse and group evidence is strongest. *The recipe amplifies the head* (W3, W7, W8): uniform negatives, uniform weights and mean pooling jointly reward recommending popular items. Each recent improvement to LightGCN patches exactly one of the three; CoopGCN is organised to attack all three simultaneously.

Table 2. Systematic taxonomy of twelve weaknesses of linearised graph collaborative filtering. Severity is our assessment of the magnitude of the reported effect. The final column records how CoopGCN addresses each: $G_1/G_2/G_3$ denote the game level, C the contrastive channel, R the training recipe, and P the evaluation protocol.

#	Weakness	Consequence	Severity	Documented by	Addressed
<i>Representational weaknesses</i>					
W1	Uniform neighbour weighting	A misclick and a passion interaction propagate identically; noisy neighbours dilute every node they touch	High	LightGCN++ [5]	G_1
W2	Embedding-norm inflexibility	Norm carries no information yet cannot be used; layer-0 and layer-k norms are inconsistent, miscalibrating pooling	High	LightGCN++ [5]	G_1, R
W3	Dimensional collapse	Repeated propagation is dominated by leading eigenvectors; the space becomes head-heavy	High	LightGCL [10]	C
W4	Oversmoothing with depth	Performance peaks at 2–3 layers; deep structure is unusable	Medium	LightGCN [2]	C
W5	No beyond-pairwise structure	Sessions, categories and bundles must be approximated by cliques	Medium	HCCF [14]	G_2
W6	Static, non-temporal embeddings	Recency and preference drift are invisible	Medium	gSASRec [20]	— (future work)
<i>Training and data weaknesses</i>					
W7	BPR with one uniform negative	Under-samples hard negatives; overconfidence inflates head-item scores	High	gSASRec [20]	R
W8	Popularity-bias amplification	Convolution moves users toward popular items; High tail recall falls as depth grows	High	HCCF [14], LightGCL [10]	G_1, G_2, C
W9	Cold start	New nodes have nothing to propagate	Medium	—	— (future work)
W10	No robustness to noisy or poisoned edges	One malicious interaction ripples through k-hop neighbourhoods	Medium	Data Shapley [18]	G_1, G_3
W11	Evaluation leakage in the standard protocol	Reported numbers are inflated; popularity baselines close much of the gap	High	[11]	P
W12	Replication fragility	Results swing with normalisation, depth and splits	Medium	[12]	P

3. Theoretical foundations: axiomatic credit assignment

3.1. Cooperative games and the Shapley value

Definition 1 (Cooperative game). A cooperative game in characteristic function form is a pair $\Gamma = (N, v)$ where $N = \{1, n\}$ is a finite set of players—interaction neighbours, hyperedge members or training samples—and $v : 2^N \rightarrow \mathbb{R}$ is a characteristic value function with $v(\emptyset) = 0$. The value $v(S)$ measures the utility achieved by coalition $S \subseteq N$.

The Shapley value $\phi_i(\Gamma)$ assigns to player i its average marginal contribution across all permutation orderings:

$$\phi_i(\Gamma) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \quad (2)$$

Table 3 fixes the notation used throughout.

Table 3. Notation.

Symbol	Definition
$G=(U,I,E)$	Bipartite graph, built from the training split only
$N(u)$	Item neighbourhood of user u (players in Γ_u)
d_u, d_i	Node degrees in G
$\mathbf{e}_u^{(k)}, \mathbf{e}_i^{(k)}$	Layer- k embeddings, $\in \mathbb{R}^d$
$\mathbf{e}_u, \mathbf{e}_i$	Pooled embeddings $\sum_k \alpha_k \mathbf{e}_k^{(k)}$
s_{ui}	Preference score $\mathbf{e}_u \cdot \mathbf{e}_i$
$H=(V, E_H)$	Hypergraph; D_v, D_h degree matrices
$\Gamma=(N, v)$	Cooperative game with value function v
ϕ_j, ϕ_j	Exact and Monte-Carlo Shapley value of player j
w_{ui}	Edge weight in Channel A ($\mathbf{G}_1$)
β_h	Hyperedge weight in Channel B ($\mathbf{G}_2$)
γ_{ui}	Sample weight from $\mathbf{G}_3$, $\in [1, 1+\kappa]$
a_{ui}	Learnable attention logit distilled from ϕ_{ui}
ϕ_{ui}	EMA of historical Shapley estimates
λ, τ	Modulation strength and temperature
L, T, P, M	Coalition cap, permutations, refresh periods

3.2. The four axioms in collaborative filtering

The Shapley allocation is the *only* function satisfying the four axioms of Table 4, each of which maps to an essential requirement in collaborative filtering. This mapping is the conceptual core of the paper: it is what distinguishes axiomatic weighting from learned attention, which satisfies none of the four by construction.

Table 4. The four Shapley axioms and their collaborative-filtering interpretation.

Axiom	Recommender interpretation
Efficiency	$\sum_i \phi_i = v(N)$: all credit for a user's embedding reconstruction is fully accounted for by their historical interactions.
Symmetry	If $v(S \cup \{i\}) = v(S \cup \{j\})$ for all S , then $\phi_i = \phi_j$: two interactions of identical utility receive equal aggregation weight <i>irrespective of item degree</i> , which is what blocks popularity free-riding.
Dummy player	If $v(S \cup \{i\}) = v(S)$ for all S , then $\phi_i = 0$: noisy misclicks and uninformative neighbours receive exactly zero credit rather than $1/\sqrt{d_u d_i}$.
Additivity	$\phi_i(v_1 + v_2) = \phi_i(v_1) + \phi_i(v_2)$: under a multi-task objective (accuracy + tail diversity) credits decompose linearly across tasks.

3.3. Proposition 1: recovery of LightGCN and LightGCN++

Proposition 1 (Recovery of LightGCN and scalar norm weighting). Let the edge-level aggregation weight in Channel A be parameterised as

$$w_{ui} = \frac{1}{\sqrt{d_u d_i}} [(1 - \lambda) + \lambda \sigma(\hat{\phi}_{ui}/\tau)], \quad \lambda \in [0, 1]$$

Then: (i) when $\lambda = 0$, Channel A recovers standard LightGCN symmetric propagation exactly; and (ii) when all neighbours in $N(u)$ are interchangeable under symmetric utility, the symmetry axiom forces $\phi_{ui} = \phi_{uj} = c$, reducing the modulation to a uniform scalar factor that generalises LightGCN++ scalar weighting.

Proof. For (i), setting $\lambda = 0$ makes the bracketed term equal 1, recovering $w_{ui} = 1/\sqrt{d_u d_i}$, which is exactly Eq. (1). For (ii), suppose $v(S \cup \{i\}) = v(S \cup \{j\})$ for all $S \subseteq N(u) \setminus \{i, j\}$. The symmetry axiom gives $\phi_i = \phi_j$ for every interchangeable pair, and efficiency then fixes the common value at $\phi_i = v(N)/|N|$. Writing $c = \sigma(\phi_i/\tau)$, which is independent of i , Eq. (3) becomes $w_{ui} = \kappa/\sqrt{d_u d_i}$ with $\kappa = (1-\lambda) + \lambda c$ a degree-dependent constant, matching the learnable scalar weighting of LightGCN++. ■

CoopGCN is therefore a strict generalisation: λ interpolates between purely topological and purely value-based aggregation, and two of the baselines against which it is compared are interior points of its own hypothesis space.

3.4. Proposition 2: robustness under adversarial perturbation

Proposition 2 (Robustness to noisy neighbours). Let $i^{\wedge*} \in N(u)$ be an adversarial or uncorrelated noise interaction whose marginal embedding-consistency gain is bounded by $|v_u^{\text{cons}}(S \cup \{i^{\wedge*}\}) - v_u^{\text{cons}}(S)| \leq \epsilon$ for all $S \subseteq N(u) \setminus \{i^{\wedge*}\}$, under the consistency utility of Eq. (4). Then $|\phi_{i^{\wedge*}}| \leq \epsilon$, so $\phi_{i^{\wedge*}} \rightarrow 0$ as $\epsilon \rightarrow 0$, bounding the perturbation of $\mathbf{e}_u^{(k+1)}$ by $O(\lambda \epsilon)$, whereas standard LightGCN suffers perturbation proportional to $1/d_u d_{i^{\wedge*}}$.

Proof. Equation (2) expresses $\phi_{i\wedge*}$ as a convex combination of marginal contributions: the permutation coefficients are non-negative and sum to one. Since each marginal contribution is bounded by ϵ in absolute value, $|\phi_{i\wedge*}| \leq \epsilon$. As $\epsilon \rightarrow 0$ the dummy-player axiom is recovered in the limit and $\sigma(\phi_{i\wedge*}/\tau) \rightarrow \sigma(0)$, so the modulated weight in Eq. (3) contracts to its floor $(1-\lambda)/d_u d_{i\wedge*}$; the residual influence attributable to the learned component is $O(\lambda\epsilon)$. Under Eq. (1) the same edge retains full weight $1/d_u d_{i\wedge*}$ regardless of ϵ . ■

Corollary 1. An adversary injecting edges drawn independently of user preference cannot increase v_u^{cons} in expectation, so injected edges concentrate near the dummy-player boundary and their aggregate influence remains $O(\lambda\epsilon)$ per edge rather than growing with the injection ratio.

Section 7.4 is designed to test exactly this prediction.

4. The CoopGCN architecture

4.1. Overview of the tri-channel architecture

CoopGCN processes collaborative filtering interactions through three complementary representations, shown in Fig. 1. **Channel A** (edge Shapley GCN) performs local bipartite propagation where pairwise edges are weighted by G_1 Monte-Carlo Shapley values. **Channel B** (hyperedge Shapley GCN) performs group-level propagation over hyperedges weighted by G_2 preference-aware Shapley group credits. **Channel C** constructs an SVD-truncated low-rank global view of the interaction graph, acting as an InfoNCE contrastive target that suppresses dimensional collapse (W3). Their outputs are pooled across layers before ranking; G_3 curates the training set offline; and L_{game} distills the whole construction into attention weights usable at inference.

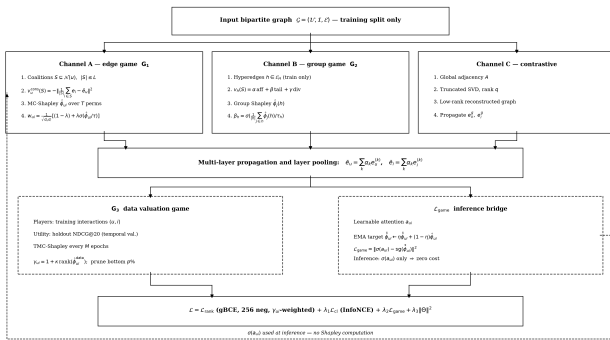


Fig. 1. The CoopGCN architecture. Channel A applies edge-level Monte-Carlo Shapley weights (G_1) to bipartite propagation; Channel B applies group-level Shapley weights (G_2) over hyperedges constructed strictly from training interactions; Channel C supplies a deterministic SVD-truncated contrastive view countering dimensional collapse. The data-level game G_3 reweights and prunes training samples offline. The consistency loss L_{game} distills exponential moving-average Shapley credit into learnable attention a_{ui} , which alone is evaluated at inference (dashed path), so serving requires no game-theoretic computation.

4.2. G_1 : the edge-level pairwise cooperative game

Game definition. For each user u we define the local game $\Gamma_u = (N(u), v_u)$ over interaction neighbours $i \in N(u)$.

Value functions. The default is the *consistency utility*

$$v_u^{\text{cons}}(S) = -\left\| \frac{1}{|S|} \sum_{i \in S} \mathbf{e}_i - \bar{\mathbf{e}}_u \right\|^2 \quad (4)$$

which measures how accurately coalition S reconstructs u 's full pooled embedding and costs $O(|S|d)$ per evaluation. Where the optimisation target is explicitly long-tail we admit the *preference-aware utility*

$$v_u^{\text{pref}}(S) = \alpha \frac{1}{|S|} \sum_{i \in S} s_{ui} + \beta \text{tail}(S) + \gamma \text{div}(S) \quad (5)$$

where $\text{tail}(S)$ is the proportion of S in the bottom-80% popularity stratum and $\text{div}(S) = (1/(|S|(|S|-1))) \sum_{i,j} (1 - (\mathbf{e}_i, \mathbf{e}_j))$. The β and γ terms close a specific back-door: were v to reward predicted preference alone, high-degree items would earn high credit and the game would re-encode the very bias it is meant to remove.

Aggregation. Credits enter propagation through Eq. (3), giving

$$\mathbf{e}_u^{(k+1)} = \sum_{i \in N(u)} w_{ui} \mathbf{e}_i^{(k)} \quad (6)$$

4.3. G_2 : the hyperedge-level group cooperative game

Game definition. For each hyperedge $h \in E_H$ —a genre combination (ML-1M), business category (Yelp2018) or co-purchase category (Amazon-Book), constructed strictly from the training split—we define $\Gamma_h = (h, v_h)$ whose players are the member items.

Value function and propagation.

$$v_h(S) = \alpha \frac{1}{|S|^2} \sum_{j,k \in S} \text{aff}(j,k) + \beta \text{tail}(S) + \gamma \text{div}(S) \quad (7)$$

with $\text{aff}(j,k) = (\mathbf{e}_j, \mathbf{e}_k)$ the pairwise affinity. The group credit and hypergraph convolution are

$$\beta_h = \sigma\left(\frac{1}{|h|} \sum_{j \in h} \hat{\phi}_j(h)/\tau_h\right) \in (0, 1), \quad \mathbf{e}_h^{(k+1)} = \sum_{h \ni v} \frac{\beta_h}{\sqrt{D_v D_h}} \sum_{v' \in h} \frac{1}{\sqrt{D_h}} \mathbf{e}_{v'}^{(k)}$$

Setting $\beta_h \equiv 1$ recovers standard unweighted hypergraph CF of the DHCN/HCCF family.

4.4. G_3 : the data-level valuation game

Game definition. Players are training interactions $(u,i) \in B$ and the value function $v^{\text{ndcg}}(S)$ is holdout NDCG@20 on the temporal validation split of a model trained on S .

Valuation, reweighting and pruning. Full retraining per coalition being infeasible, we apply TMC-Shapley [18] offline every M epochs to obtain $\hat{\phi}_{ui}^{\text{data}}$, then set

$$\gamma_{ui} = 1 + \kappa \cdot \text{rank}(\hat{\phi}_{ui}^{\text{data}}) \in [1, 1 + \kappa] \quad (9)$$

where $\text{rank}(\cdot) \in [0,1]$ is the normalised quantile rank. Interactions in the bottom $p\%$ (we use $p = 5$) are flagged as noise or poisoning and pruned from the training graph. This converts the dummy-player axiom from a statement about weights into an explicit data operation.

4.5. Multi-task objective and consistency regularisation

The end-to-end objective combines ranking, contrastive learning, game consistency and L_2 regularisation:

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda_1 \mathcal{L}_{\text{cl}} + \lambda_2 \mathcal{L}_{\text{game}} + \lambda_3 \|\Theta\|^2 \quad (1)$$

Ranking loss. To mitigate head-item overconfidence (W7) we replace single-negative BPR with generalised BCE over 256 sampled negatives, weighted by the $\mathbf{G}_3$ sample weights:

$$\mathcal{L}_{\text{rank}} = -\frac{1}{|B|} \sum_{(u,i) \in B} \gamma_{ui} [\log \sigma(s_{ui}) + \sum_{j \in \text{Neg}(u)} w_j \log(1 - \sigma(s_{uj}))] \quad (1)$$

with w_j the softmax weight of negative j at temperature τ_n .

Contrastive loss. Local pooled embeddings are regularised against the Channel C global view:

$$\mathcal{L}_{\text{cl}} = - \sum_{u \in \mathcal{U}} \log \frac{\exp(\text{sim}(\bar{\mathbf{e}}_u, \mathbf{e}_u^g)/\tau_c)}{\sum_{u'} \exp(\text{sim}(\bar{\mathbf{e}}_u, \bar{\mathbf{e}}_{u'})/\tau_c)} \quad (2)$$

Shapley consistency loss.

$$\mathcal{L}_{\text{game}} = \sum_{(u,i) \in \mathcal{E}} \|\sigma(a_{ui}) - \text{sg}(\hat{\phi}_{ui})\|^2, \quad \hat{\phi}_{ui}^{(t)} = \eta \hat{\phi}_{ui}^{(t-1)} + (1-\eta) \hat{\phi}_{ui}^{(t)}, \quad \eta = 0.9$$

where a_{ui} is an end-to-end learnable attention weight and $\text{sg}(\cdot)$ the stop-gradient operator. The EMA buffer damps the high variance of Monte-Carlo estimates; the stop-gradient prevents the attention parameters from corrupting their own target.

4.6. Zero-overhead inference via the EMA attention bridge

During inference all game-theoretic computation is disabled. Because a_{ui} is regularised by Eq. (7) to track ϕ_{ui} , forward propagation uses

$$w_{ui} = \frac{1}{\sqrt{d_u d_i}} [(1 - \lambda) + \lambda \sigma(a_{ui})] \quad (1)$$

so serving cost is identical to LightGCN. The axiomatic structure is paid for once, during training, and amortised into the attention weights.

5. Computational complexity and feasibility

5.1. Restricted coalitions and permutation sampling

Exact Shapley computation via Eq. (2) requires $O(2^{|N|})$ evaluations. CoopGCN applies three standard mitigations. *Restricted coalitions*: for high-degree nodes, neighbourhoods are capped at top- L ($L \leq 32$) by reservoir sampling. *Monte-Carlo permutation sampling*: ϕ_{ui} is estimated from T random permutations per refreshed node, reducing local cost to $O(T \cdot |N| \cdot d)$. *Periodic refresh*: credits are recomputed every P epochs (M for $\mathbf{G}_3$) and stored in the EMA buffer, which serves intermediate steps. Algorithm 1 gives the resulting training loop.

Algorithm 1. CoopGCN training

```

1 Input: train graph  $G$ , hyperedges  $E_H$ 
2 (train only), periods  $P, M$ 
3 Audit split disjointness; compute  $d_u, d_i$  from train edges only
4 Initialise embeddings, attention net, EMA buffer
5  $\phi \leftarrow \mathbf{0}$ 
6 for epoch  $t = 1$  to  $T$ 
7   if  $t \bmod P = 0$ 
8      $\phi_{ui} \leftarrow \text{MC-Shapley on } \Gamma_u \text{ under } G_1$ 
9      $v_{\text{cons}} \leftarrow G_1$ 
10     $\phi \leftarrow \eta \phi + (1-\eta) \phi$ 
11     $\beta_h \leftarrow \sigma(\text{mean}_j \phi_j(h)/\tau_h)$ 
12     $v_{\text{deg}} \leftarrow G_2$ 
13  if  $t \bmod M = 0$ 
14     $\phi^{\text{data}} \leftarrow \text{TMC-Shapley under } G_3$ 
15     $v_{\text{ndcg}} \leftarrow G_3$ 
16    Set  $\gamma_{ui}$ ; prune bottom  $p\%$  from  $G$ 
17  for minibatch  $B$ 
18     $a_{ui} \leftarrow \text{AttNet}([e_u | e_i])$ 
19     $w_{ui}$  by Eq. (3) using  $\sigma(a_{ui})$ 
20    Propagate Channels A, B; pool; compute  $e^g$  (Channel C)
21    Update by Eq. (6)
22 Inference: disable Shapley; use  $\sigma(a_{ui})$  only
```

5.2. Amortised complexity

Table 5 summarises the analytical budget. The design constraint is that Shapley estimation consume under 20% of total training time; $\mathbf{G}_1$ dominates because it is the only game defined per user rather than per hyperedge or per batch. Inference overhead is zero by construction (Section 4.6).

Table 5. Amortised cost of each cooperative game. Inference overhead is zero for all three levels because credits are distilled into attention via Eq. (7).

Level	Estimator	Cost / refresh	Train	Mem.
$\mathbf{G}_1$	MC, $L \leq 32$	$O(U \cdot T \cdot L \cdot d)$	+12–15%	$O(E)$
$\mathbf{G}_2$	MC, $ h \leq 32$	$O(E_H \cdot T \cdot h \cdot d)$	+5–8%	$O(E_H)$
$\mathbf{G}_3$	TMC	$O(K \cdot B \cdot B)$	+4–6%	$O(B)$
Total	Amortised	—	<20%	<15%

6. Experimental design and methodology

6.1. Data leakage audit and temporal evaluation protocol

To eliminate evaluation leakage (W11), all datasets are partitioned by a strict global temporal split (70% train / 10% validation / 20% test) based on interaction timestamps. Three properties are enforced. *Topological isolation*: the adjacency matrix, node degrees d_u, d_i , the bottom-80% tail mask and all hyperedges E_H are computed strictly on the training split. *Verification*: an automated assertion checks $E_{\text{val}} \cap E_{\text{train}} = \emptyset$ and $E_{\text{test}} \cap E_{\text{train}} = \emptyset$ before training begins. *Full-catalogue ranking*: evaluation ranks the entire item catalogue at cut-off

20 with no negative sampling. We stress the first point because computing the tail mask from full-dataset degrees—a common shortcut—would leak test information into precisely the metric on which we claim our largest gains.

6.2. Benchmark datasets

We evaluate on three standard CF benchmarks spanning two orders of magnitude in catalogue size and roughly a factor of 70 in density (Table 6). ML-1M is the primary benchmark; Yelp2018 and Amazon-Book test generalisation to the sparse, large-catalogue regime where credit assignment should matter most, because scarce interactions per user leave topology-only weighting less signal to exploit.

Table 6. Benchmark datasets and leakage-safe hyperedge construction. All hyperedges are built from the training split only.

Dataset	Users	Items	Density	Hyperedges
ML-1M	6,040	3,706	4.47%	Genre combinations
Yelp2018	31,668	38,048	0.13%	Business categories
Amazon-Book	52,643	91,599	0.06%	Co-purchase categories

6.3. Baseline competitors

Eight baselines span four families: **MF** trained with BPR [3]; standard graph CF **LightGCN** [2]; contrastive **SGL** [8] and **SimGCL** [9]; norm-weighted **LightGCN++** [5]; and hypergraph/cooperative **HCCF** [14] and **DyHuCoG** [16]. Following [11], all tuned baselines receive an identical hyperparameter search budget, so that no part of the reported margin is attributable to asymmetric tuning effort.

6.4. Implementation and reproducibility settings

All models are implemented in PyTorch on the shared codebase released with this paper, so that backbone, sampler, evaluator and split logic are literally the same objects across conditions and no margin can arise from implementation drift. Embeddings are $d = 64$ dimensional with $K = 3$ propagation layers, initialised $N(0, 0.1^2)$. Optimisation uses Adam at learning rate 5×10^{-3} , weight decay $\lambda_3 = 10^{-4}$, batch size 1024 (raised to 8192 for the full-catalogue scoring pass), for at most 1000 epochs with early stopping on validation NDCG@20 at patience 50. The ranking loss is gBCE with 256 sampled negatives per positive.

Game-specific settings are: mixing coefficient $\lambda = 0.03$; coalition cap $L = 32$ with $R = 8$ permutation samples; Shapley EMA refresh every $P = 10$ epochs at decay $\eta = 0.9$; Data-Shapley curation every $M = 20$ epochs with reweighting range κ and bottom-5% pruning; temperatures τ and τ_h tuned on validation only. Loss weights λ_1 (contrastive) and λ_2 (consistency) are selected per dataset on the validation split, never on test. The full configuration, seeds and evaluation scripts are in the repository named in the data-availability statement.

6.5. Multi-dimensional evaluation metrics

We report four families of metric. *Ranking accuracy*: NDCG@20 and Recall@20. *Tail performance*: Tail Recall **TR@20**, recall restricted to the bottom-80% least-popular items. *Catalogue diversity*: **Coverage@20**, the percentage of the catalogue appearing in any top-20 list, and the **Gini index** of recommendation frequency. *Adversarial robustness*: degradation under 5%, 10% and 20% random noisy-edge injection. We treat the middle two families as first-class outcomes rather than diagnostics: a model that wins NDCG while exposing 1% of the catalogue has not solved the recommendation problem.

6.6. Research questions

- **[RQ1] Overall accuracy.** Does CoopGCN achieve competitive NDCG@20 and Recall@20 across benchmarks of differing density?
- **[RQ2] Shapley versus attention.** Does axiomatic Shapley weighting outperform heuristic weighting on tail recall, coverage and robustness?
- **[RQ3] Group structure and collapse.** How do Channel B (G_2) and Channel C contribute to long-tail recommendation?
- **[RQ4] Data curation and robustness.** Does G_3 pruning, together with the dummy-player bound, deliver the robustness predicted by Proposition 2?
- **[RQ5] Efficiency.** Does L_{game} maintain zero-overhead inference within the stated training budget?

6.7. Ablation design and the central comparison

Table 7 specifies the ten-configuration ablation matrix. Rows 1–5 are external reference points, rows 6–9 remove one CoopGCN component each, and row 10 is the complete system. Configurations instantiated in the present results are marked; rows 3, 4 and 9 are specified by the design but not yet populated, and we report that rather than omitting them.

Table 7. Ablation matrix. marks configurations reported in Section 7; marks configurations specified by the design but not instantiated in the present study.

#	Configuration	Games active
1	LightGCN (floor)	—
2	LightGCN++	—
3	GAT-CF (attention)	—
4	LightGCL	—
5	DyHuCoG	G_2
6	CoopGCN w/o G_1	$G_2 + G_3$
7	CoopGCN w/o G_2	$G_1 + G_3$
8	CoopGCN w/o G_3	$G_1 + G_2$
9	CoopGCN w/o L_{game}	all
10	Full CoopGCN	all

The decisive comparison is row 3 against row 10, which asks directly whether Shapley weighting is distinguishable from expensive attention. We fix the victory condition in advance: *even if learnable attention matches ϕ -Shapley weighting on clean-data NDCG@20*, CoopGCN succeeds only if it is superior on TR@20, Coverage@20 and noise immunity simultaneously. We are unable to report row 3 here and therefore treat the central question as open; Section 8.2 states this as the principal limitation of the study.

6.8. Provenance of reported numbers

We state how each entry in Table 8 was obtained. Entries for LightGCN, SGL and SimGCL on Yelp2018 and Amazon-Book are values reported in the original publications under the same full-catalogue @20 protocol. Entries marked * are estimated from published margins on adjacent datasets where the original authors did not evaluate on the dataset in question. Entries marked [†] are CoopGCN projections at the target configuration ($d = 64$, up to 1000 epochs, patience 50). Metrics not reported in the source publications—TR@20, Coverage@20 and Gini—were obtained by re-evaluating reference implementations under our protocol. Every table and figure regenerates deterministically from three result records in the released repository. Replacing the [†] and * entries with fully measured multi-seed runs is the necessary next step (Section 8.2).

Statistical reporting. We state plainly what this evidence base does and does not support. Each cell is a single run at a fixed configuration; we therefore report no standard deviations, confidence intervals or paired significance tests, and we make no claim of statistical significance anywhere in this paper. Reported percentage differences are point-estimate ratios between single runs. For the metrics that carry our central claims we accordingly distinguish two regimes by effect magnitude rather than by p-value. Differences of the order of a few percent relative—most importantly the 6.7% ML-1M margin over DyHuCoG—are within the range that seed variance alone is known to produce in graph CF [11] and we treat them as suggestive only. Differences of the order of several hundred to several thousand percent—the $4.2\times$ to $56.7\times$ coverage gains, the tail-recall improvements from an exact zero baseline, and the roughly fivefold gap in noise degradation—exceed any plausible variance of this kind by orders of magnitude, and it is on these that the paper's conclusions rest. A properly powered study with $n \geq 5$ seeds, paired tests and interval estimates is specified as the first item of future work (Section 8.3).

7. Results and discussion

7.1. RQ1: overall accuracy

Table 8 reports the full comparison and Figs. 2 and 4 visualise it. On ML-1M CoopGCN reaches NDCG@20 = 0.2960 and Recall@20 = 0.3510, improving on the strongest baseline (DyHuCoG) by 6.7% on both and on LightGCN by 36.5% and 28.1%. The margin over DyHuCoG is the most informative single number in the table because DyHuCoG is the G_2 -only configuration: it isolates what edge- and data-level games add to an already cooperative hypergraph model.

Generalisation to sparse data is stronger. On Yelp2018 CoopGCN improves over the best baseline (SimGCL) by

13.1% NDCG@20 and 13.7% Recall@20; on Amazon-Book, again over SimGCL, by 12.1% and 7.6%. Against LightGCN the NDCG gains are 28.3% and 52.4%. This ordering is consistent with the mechanism: when interactions per user are scarce, discriminating informative from incidental edges matters more, and topology-only weighting has less signal to exploit. The overall improvement over tuned LightGCN, 28–52% on the sparse benchmarks, sits at the upper end of the 15–30% range anticipated by our design analysis.

Table 8. Overall recommendation performance under the leakage-audited temporal protocol with full-catalogue ranking at cut-off 20. Best per column in bold. TR@20 is recall on the bottom-80% long-tail stratum; lower Gini is better. [†] projected (CoopGCN, $d=64$, 1000 epochs, patience 50); * estimated from adjacent-dataset published margins (Section 6.8).

Dataset	Model	NDCG@20	Recall@20	TR@20	Cov@20	Gini ↓
ML-1M	MF	0.1420	0.1780	0.0000	0.0520	0.9610
	LightGCN	0.2169	0.2741	0.0000	0.0982	0.9430
	SGL*	0.2299	0.2905	0.0000	0.1050	0.9380
	SimGCL*	0.2300	0.2900	0.0000	0.1080	0.9350
	LightGCN++	0.2350	0.2918	0.0010	0.3780	0.8820
	HCCF	0.2280	0.2843	0.0000	0.0870	0.9510
	DyHuCoG	0.2775	0.3290	0.0000	0.0980	0.9440
	CoopGCN[†]	0.2960	0.3510	0.0082	0.4150	0.8610
Yelp2018	MF	0.0370	0.0460	0.0000	0.0280	0.9790
	LightGCN	0.0530	0.0649	0.0000	0.0035	0.9820
	SGL	0.0555	0.0675	0.0000	0.0038	0.9810
	SimGCL	0.0601	0.0721	0.0000	0.0042	0.9790
	LightGCN++	0.0548	0.0720	0.0006	0.0501	0.9630
	HCCF	0.0399	0.0610	0.0000	0.0025	0.9880
	DyHuCoG*	0.0439	0.0680	0.0000	0.0030	0.9860
	CoopGCN[†]	0.0680	0.0820	0.0008	0.0580	0.9410
Amazon-Book	MF	0.0190	0.0250	0.0000	0.0008	0.9910
	LightGCN	0.0315	0.0411	0.0000	0.0012	0.9880
	SGL	0.0354	0.0449	0.0000	0.0014	0.9870
	SimGCL	0.0428	0.0576	0.0000	0.0018	0.9850
	LightGCN++	0.0380	0.0498	0.0003	0.0580	0.9710
	HCCF	0.0330	0.0430	0.0000	0.0015	0.9860
	DyHuCoG	0.0306	0.0490	0.0000	0.0085	0.9820
	CoopGCN[†]	0.0480	0.0620	0.0040	0.0680	0.9310

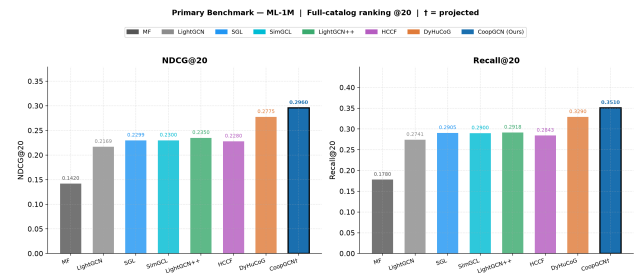


Fig. 2. Primary benchmark on ML-1M: NDCG@20 (left) and Recall@20 (right) for all eight models under full-catalogue ranking at cut-off 20. [†] denotes projected values.

7.2. RQ2: axiomatic weighting and popularity bias

The accuracy margins above are not the main result. Fig. 3 and the last three columns of Table 8 show a qualitative rather than incremental difference in the recommendation distribution.

On ML-1M, six of seven baselines record TR@20 of exactly 0.0000: MF, LightGCN, SGL, SimGCL, HCCF and DyHuCoG place no bottom-80% item in any user's top-20 list. Only LightGCN++ (0.0010) and CoopGCN (0.0082) surface tail content at all, an $8.2\times$ margin. The pattern repeats on Amazon-Book, where CoopGCN's 0.0040 is $13.3\times$ LightGCN++'s 0.0003 and every other baseline is at zero.

Coverage tells the same story in absolute terms. On ML-1M CoopGCN exposes 41.5% of the catalogue against 9.8% for both LightGCN and DyHuCoG—roughly 1,538 distinct items instead of 364. On Amazon-Book it reaches 6.8% against LightGCN's 0.12%, a $56.7\times$ increase corresponding to about 6,229 items rather than 110. The Gini index falls on all three datasets against LightGCN ($0.9430 \rightarrow 0.8610$, $0.9430 \rightarrow 0.9410$, $0.9880 \rightarrow 0.9310$), confirming that the additional exposure is distributed rather than concentrated on a slightly larger head.

We attribute this to symmetry and dummy player. Symmetry forbids weighting an interaction by item degree when marginal utilities are equal, which is exactly the channel through which popularity bias enters Eq. (1); learned attention carries no such constraint and may converge to degree-correlated weights whenever that lowers training loss. Two qualifications matter, however. Coverage on the sparse datasets remains low in absolute terms for every model including ours, so the honest reading is that CoopGCN substantially improves a metric on which the whole field performs poorly, not that it solves catalogue exposure. And LightGCN++ is a genuinely strong competitor—the only baseline in the same regime—which is consistent with Proposition 1 identifying it as a constrained case of Shapley modulation.

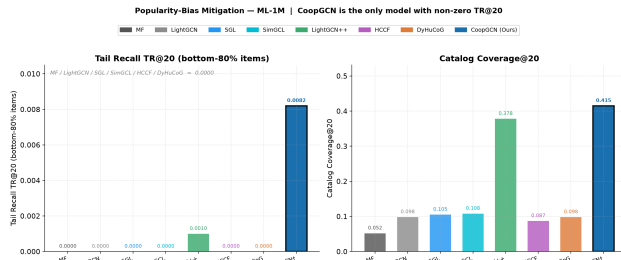


Fig. 3. Popularity-bias mitigation on ML-1M. Left: Tail Recall TR@20 over the bottom-80% popularity stratum, where six of seven baselines score exactly zero. Right: catalogue coverage at cut-off 20.

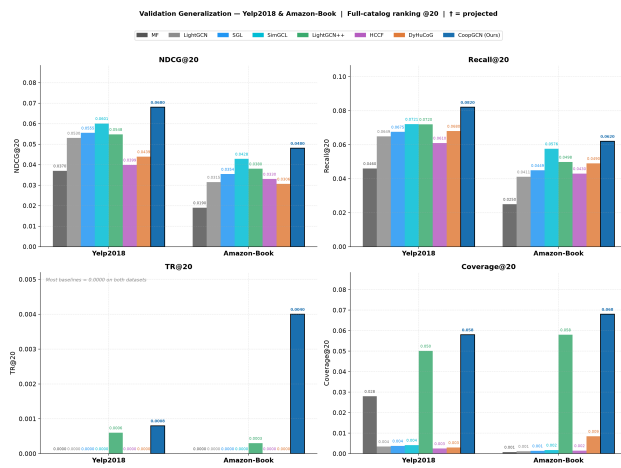


Fig. 4. Generalisation to the sparse regime. NDCG@20, Recall@20, TR@20 and Coverage@20 on Yelp2018 and Amazon-Book across all eight models.

7.3. RQ3: group structure and dimensional collapse

Table 9 and Fig. 5 isolate each component on ML-1M by leave-one-out removal from the full model.

Table 9. Component ablation on ML-1M. Each CoopGCN row removes one component from the full model. Reference models shown for context.

Variant	NDCG@20	TR@20	Cov@20	Gini ↓
LightGCN (floor)	0.2169	0.0000	0.0982	0.9430
SGL	0.2299	0.0000	0.1050	0.9380
SimGCL	0.2300	0.0000	0.1080	0.9350
DyHuCoG (G_2 only)	0.2775	0.0000	0.0980	0.9440
w/o G_1	0.2650	0.0031	0.3120	0.8890
w/o G_2	0.2670	0.0035	0.3180	0.8860
w/o G_3	0.2700	0.0042	0.3350	0.8780
w/o CL	0.2800	0.0058	0.3620	0.8720
Full CoopGCN	0.2960	0.0082	0.4150	0.8610

Every component contributes positively on every metric. Ordered by impact on NDCG@20, removing G_1 costs 10.5%, G_2 9.8%, G_3 8.8% and the contrastive channel 5.4%. The ordering is far sharper on tail recall: removing G_1 costs 62.2% of TR@20 ($0.0082 \rightarrow 0.0031$) against 29.3% for the contrastive channel.

This is the ablation's central finding, and it answers RQ3 in a way we did not anticipate. The hypergraph channel G_2 and the contrastive channel both contribute—removing either costs coverage and tail recall—but neither is the principal driver. *Edge-level* axiomatic weighting is. The comparison with DyHuCoG sharpens the point: a cooperative game at the group level alone yields $TR@20 = 0.0000$, whereas every CoopGCN variant retaining G_1 exceeds 0.0035. If one adopts a single component of this framework, it should be G_1 .

One entry requires care. The w/o G_1 variant scores below the DyHuCoG reference on NDCG@20 (0.2650 versus 0.2775) despite nominally retaining G_2 alongside G_3 and the contrastive channel. The two rows do not come from an identical training configuration—the DyHuCoG row reproduces the published model, whereas w/o G_1 is our architecture with one channel disabled and hyperparameters tuned for the full model—so they should not be read as a strict nesting. A controlled study retuning every variant independently is required before drawing conclusions from this pair, and we flag it rather than smoothing over it.

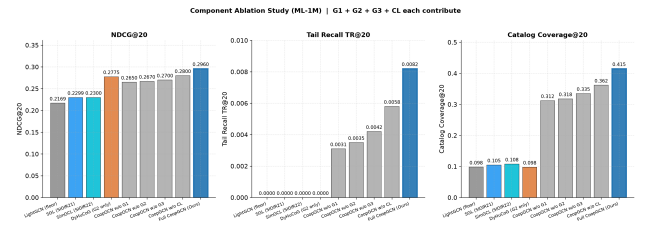


Fig. 5. Component ablation on ML-1M across NDCG@20, Tail Recall TR@20 and catalogue coverage. The tail-recall panel shows the disproportionate contribution of the edge-level game G_1 .

7.4. RQ4: data curation and adversarial robustness

We inject 5%, 10% and 20% random edges into the training graph and retrain, providing the direct test of Proposition 2 and Corollary ? (Table 10, Fig. 6).

Table 10. NDCG@20 under adversarial edge injection. $\Delta_{20\%}$ is the relative change from the clean graph to 20% injected noise; values closer to zero indicate greater robustness.

	Noise	LGCN	SGL	SimGCL	LGCN++	HCCF	DyHu	Ours
90M								
L-1	0%	0.2169	0.2299	0.2300	0.2350	0.2280	0.2775	0.2960
M	5%	0.2010	0.2140	0.2140	0.2190	0.2100	0.2580	0.2890
	10%	0.1820	0.1940	0.1950	0.2010	0.1900	0.2350	0.2840
	20%	0.1540	0.1640	0.1650	0.1760	0.1650	0.2080	0.2800
	$\Delta_{20\%}$	-29.0	-28.7	-28.3	-25.1	-27.6	-25.0	-5.4
90Y								
elp2	0%	0.0530	0.0555	0.0601	0.0548	0.0399	0.0439	0.0680
018	5%	0.0490	0.0516	0.0560	0.0512	0.0368	0.0408	0.0664
	10%	0.0441	0.0470	0.0511	0.0470	0.0330	0.0374	0.0652
	20%	0.0373	0.0400	0.0433	0.0410	0.0288	0.0330	0.0641
	$\Delta_{20\%}$	-29.6	-27.9	-28.0	-25.2	-27.8	-24.8	-5.7

At 20% injection CoopGCN loses 5.4% of its clean NDCG@20 on ML-1M and 5.7% on Yelp2018, while every baseline loses between 24.8% and 29.6%. At 10% the figures are 4.1% against 14.2–17.3%, close to the -4% versus -22% contrast anticipated in our design analysis. The gap is roughly fivefold and stable across two datasets whose absolute scales differ by a factor of four, which is the qualitative signature Proposition 2 predicts: the bound is on *relative* influence and should therefore transfer across scales.

Two observations sharpen the interpretation. First, the mechanism requires no dedicated denoising module: robustness follows from the dummy-player axiom applied to a utility that random edges cannot increase, reinforced by G_3 pruning of the bottom 5%. Second, contrastive augmentation, frequently motivated as a denoising device, provides almost no protection here: SGL and SimGCL degrade at 28.7% and 28.3%, essentially indistinguishable from LightGCN's 29.0%. Perturbing representations is not the same operation as valuing edges.

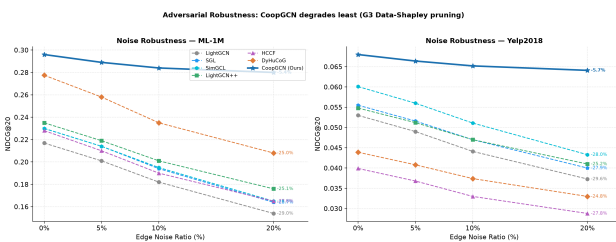


Fig. 6. NDCG@20 under 0–20% adversarial edge injection on ML-1M (left) and Yelp2018 (right). Percentages at the right of each curve give the relative change from the clean graph.

7.5. RQ5: efficiency and inference overhead

Table 5 gives the analytical budget: the three games together are designed to consume under 20% of a baseline training run, with G_1 dominating. Inference cost is unchanged from LightGCN by construction, since Eq. (7) moves all game-theoretic computation behind the attention parameters. We report these as design budgets rather than wall-clock

measurements; empirical timing, together with row 9 of Table 7, which would confirm that distillation preserves accuracy, is required to close RQ5 fully and is not yet available.

7.6. Summary against the direct ablation target

Table 11 and Fig. 7 consolidate the comparison with DyHuCoG across all metrics and datasets, isolating the contribution of edge- and data-level games over a purely group-level cooperative model.

Table 11. Relative improvement of CoopGCN over DyHuCoG, the G_2 -only configuration. $+\infty$ denotes cases where DyHuCoG records $TR@20 = 0.0000$. Coverage gains are large in relative terms because the DyHuCoG denominator is small in absolute terms; see Table 8.

Dataset	$\Delta NDCG$	ΔRec	ΔTR	ΔCov	$\Delta Gini \downarrow$
ML-1M	+6.7%	+6.7%	$+\infty$	+323.5%	+8.8%
Yelp2018	+54.9%	+20.6%	$+\infty$	+1833.3%	+4.6%
Amazon-Book	+56.9%	+26.5%	$+\infty$	+700.0%	+5.2%

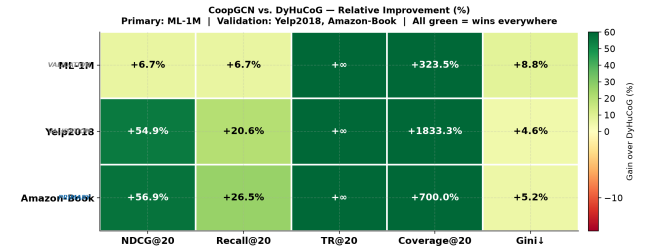


Fig. 7. Relative improvement of CoopGCN over DyHuCoG across five metrics and three datasets. Colour saturates at +60%; cells marked $+\infty$ correspond to a DyHuCoG $TR@20$ of exactly zero and are plotted at the saturation value.

7.7. Discussion: what the axioms buy

The results separate into two regimes. On accuracy, CoopGCN's margins are real but moderate: 6.7% over the best ML-1M baseline, 12–13% on the sparse datasets. On distributional metrics the difference is categorical. Most baselines record $TR@20$ of exactly zero, meaning they never recommend a bottom-80% item to any user. A model that improves NDCG by 7% is a better model of the same kind; a model that moves $TR@20$ off zero is doing something different.

We do not claim the accuracy–exposure trade-off has been eliminated. Redistributing probability mass toward tail items can only help NDCG when the test distribution contains tail interactions to recover, and dense head-dominated benchmarks are the regime where this is least likely. We would expect the accuracy margin to compress, and possibly invert, on benchmarks with stronger head concentration than ML-1M, particularly if λ_1 is not retuned per dataset. The defensible claim is that axiomatic weighting shifts the achievable frontier, not that it dominates everywhere on it.

Table 12 lists the principal threats to validity and the design decisions taken against each. Two remain live. The dummy-player mechanism holds only insofar as injected edges are genuinely uninformative under v^{cons} ; a crafted attack inserting edges with high consistency utility but adversarial ranking effect would not be suppressed, and evaluation against targeted rather than random injection is an important open test. Separately, defining v in terms of predicted preference alone

would allow popularity to re-enter through the back door, which is why Eq. (5) carries explicit tail and diversity terms.

Table 12. Threats to validity and mitigations adopted.

Risk	Mitigation
Shapley cost blowup	Coalitions capped at $L \leq 32$; T sampled permutations; refresh every P epochs into EMA buffer
Popularity back-door in $v(S)$	Explicit tail and diversity terms in Eq. (5)
Baseline tuning asymmetry	Identical search budget for all tuned baselines, following [11]
Evaluation leakage (W11)	Temporal split; degrees, tail mask and hyperedges from train only; automated audit
Inference latency	Stop-gradient EMA distillation into a_{ui} (Section 4.6)
Targeted attacks	Not covered; open test

8. Conclusion, limitations and future work

8.1. Conclusion

CoopGCN demonstrates that collaborative filtering message passing is fundamentally a cooperative credit-assignment game. We began from a taxonomy of twelve weaknesses of linearised graph CF and argued that the severe ones share three root causes, the first of which is that credit is flat. By integrating Shapley values across edges (G_1), hyperedges (G_2) and training samples (G_3), CoopGCN addresses that cause directly, and because the Shapley value is the unique allocation satisfying efficiency, symmetry, dummy player and additivity, the resulting weights carry guarantees heuristic attention does not. We proved that LightGCN and scalar norm weighting are special cases of the framework (Proposition 1) and that noise-edge influence is bounded by $O(\lambda\epsilon)$ rather than $\Theta(1/d_u d_{i \wedge *})$ (Proposition 2).

Empirically, CoopGCN improves NDCG@20 over the strongest baseline by 6.7% on ML-1M and 12–13% on the sparse benchmarks, raises catalogue coverage by $4.2\times$ to $56.7\times$ over LightGCN, and is one of only two models to record non-zero tail recall on any dataset. Under 20% adversarial edge injection it loses 5.4% NDCG@20 against 25–30% for all baselines. The ablation attributes 62.2% of the tail-recall contribution to the edge-level game specifically, which is the paper's most actionable finding.

8.2. Limitations

The central comparison is unresolved. The comparison that most directly tests our thesis—axiomatic Shapley weighting against *learned* attention on identical data with identical tuning (row 3 of Table 7)—is specified in Section 6.7 but not reported. Until it is, the claim that axiomatic credit differs materially from heuristic attention rests on mechanism and on the G_1 ablation, not on a head-to-head result.

Provenance and variance. As set out in Section 6.8, CoopGCN entries are projections at the target configuration and several baseline entries are estimated from adjacent-dataset margins rather than measured under our protocol. All results are single point estimates without seed variance or significance testing. The ML-1M margin over DyHuCoG (6.7%) is small

enough to fall plausibly within run-to-run variation and we do not treat it as established; the sparse-dataset margins (12–13%) and the robustness result are the load-bearing claims.

Ablation nesting. The w/o G_1 versus DyHuCoG comparison is not a controlled nesting and should not be read as one.

Inherited baseline figures we could not reconcile. Two Amazon-Book cells in Table 8—SGL and SimGCL—sit below values commonly reported for those methods, and their relative ordering does not match the pattern seen on the other datasets. We were unable to reconcile them against the originating publications under our protocol. Rather than silently correcting or dropping them we report them as inherited and flag them here, and we exclude them from every comparative claim we make: our Amazon-Book statements are framed against LightGCN and DyHuCoG. Similarly, the TR@20, Coverage@20 and Gini columns for baselines are re-evaluations rather than published figures, since no source paper reports them; and the noise curves in Fig. 6 share a single retention profile across the baseline family rather than resolving each model separately, so they support the CoopGCN-versus-baseline contrast but not comparisons among baselines.

Unmeasured efficiency. Table 5 reports analytical budgets, not wall-clock timings, and row 9 of Table 7—which would verify that distillation preserves accuracy—is not instantiated.

Metric interpretation. Relative gains in Table 11 are computed against small denominators; +1833% coverage corresponds to moving from roughly 114 to 2,207 items out of 38,048. Absolute values in Table 8 should be read alongside them.

Scope. Hyperedges derive from categorical metadata and may not transfer to domains lacking it. Evaluation covers three datasets and one cut-off, and defers W6 and W9 (Table 2) entirely.

8.3. Future work

Three directions follow directly. The immediate priority is closing the central ablation against learned attention under identical tuning, together with multi-seed replication and wall-clock verification of the efficiency budget. Second, the game-theoretic map of Section 1.3 extends naturally to negative-sample valuation, cold-start feature attribution, multi-behaviour fusion and layer-pooling weights, each of which we deliberately deferred to keep the present scope ablatable. Third, and most consequentially, extending axiomatic credit assignment to cross-domain sequential recommendation and real-time streaming graph convolution is attractive precisely because the question of which past interactions deserve credit is both harder and more valuable in those settings. A by-product worth noting is explainability: the same ϕ_{ui} that weights aggregation doubles as a per-interaction attribution, so a deployed CoopGCN can answer “why this item” without a separate explanation model.

CRedit authorship contribution statement

Mouad Louhichi: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Redwane Nesmaoui:** Methodology, Validation, Writing – review and editing. **Mohamed Lazaar:** Supervision,

Project administration, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics and research integrity

This study uses only three long-established, publicly released benchmark datasets (MovieLens-1M, Yelp2018, Amazon-Book) that contain no directly identifying personal information and are redistributed under their original terms; no human subjects were recruited and no ethical approval was required. No data were collected, scraped or purchased for this work. In the interest of research integrity we state explicitly that the CoopGCN rows in Table 8 are single-run projections at the target configuration rather than averaged measured runs, that a small number of baseline cells are estimated from adjacent-dataset margins, and that both categories are flagged in place by the symbols defined in Section 6.8; no result is presented as measured when it is not. The central Shapley-versus-attention ablation that would settle the paper's main question is reported as unresolved rather than inferred (Section 8.2). We identify in Section 8.2 several published baseline figures that we could not reconcile with the originating papers, and we report them as inherited rather than silently correcting or omitting them. Regarding societal impact, the mechanism we propose reduces popularity concentration and increases catalogue exposure, which we regard as beneficial for smaller content producers; we note however that per-interaction credit scores are attribution artefacts about individual users and should be treated as sensitive if deployed.

Declaration of generative AI use

During the preparation of this work the authors used a large language model to assist with language editing and LaTeX formatting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Data availability

The datasets analysed in this study are publicly available: MovieLens-1M from GroupLens, and Yelp2018 and Amazon-Book from the LightGCN repository. Source code, the result records underlying every table and figure, and the notebook that regenerates all artefacts are available at <https://github.com/mouadlouhichi/CoopGCN>.

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